TSUNAMI AND COASTAL HAZARD MANAGEMENT

TSUNAMI GENERATION CHAIN → COASTAL-HAZARD FIREWALL → SOURCE-TO-IMPACT MATRIX → DETECTION-CONFIRMATION RAIL → ITEWC AUTHORITY MAP → WARNING-UNCERTAINTY LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◇ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 TSUNAMI GENERATION CHAIN

TSUNAMI GENERATION CHAIN · SUDDEN WATER DISPLACEMENT · VOLCANIC ACTIVITY · LONG WAVES · COASTAL TRANSFORMATION · RUN-UP + INUNDATION + RETURN FLOW

RISK CONDITION / INPUT
- SUDDEN WATER DISPLACEMENT
- VOLCANIC ACTIVITY

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- LONG WAVES → COASTAL TRANSFORMATION
- RUN-UP + INUNDATION + RETURN FLOW

LAST-MILE / INCLUSION / OUTCOME LIMIT
- SUDDEN WATER DISPLACEMENT
- VOLCANIC ACTIVITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LONG WAVES → COASTAL TRANSFORMATION; RUN-UP + INUNDATION + RETURN FLOW.

01

STAGE 01 COASTAL-HAZARD FIREWALL

COASTAL-HAZARD FIREWALL · SUDDEN DISPLACEMENT · GRAVITATIONAL CYCLE · STORM SURGE · CYCLONE WIND + PRESSURE · DIFFERENT MONITORING AND RESPONSE LOGIC

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
TSUNAMI → sudden displacement	STORM SURGE → cyclone wind + pressure	TSUNAMI → sudden displacement
TIDE → gravitational cycle	RULE → different monitoring and response logic	TIDE → gravitational cycle

HAZARD / ACTOR / INSTRUMENT
- TSUNAMI → sudden displacement
- TIDE → gravitational cycle

DECISIVE RISK OR MANDATE DISTINCTION
- STORM SURGE → cyclone wind + pressure
- RULE → different monitoring and response logic

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- TSUNAMI → sudden displacement
- TIDE → gravitational cycle

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORM SURGE → cyclone wind + pressure; RULE → different monitoring and response logic.

02

STAGE 02 SOURCE-TO-IMPACT MATRIX

SOURCE-TO-IMPACT MATRIX · MECHANISM DEPTH DISPLACEMENT · DISTANCE AND OCEAN CONFIGURATION · BATHYMETRY TOPOGRAPHY LAND USE · EXPOSURE VULNERABILITY AND ACTION

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
SOURCE → mechanism depth displacement	COAST → bathymetry topography land use	SOURCE → mechanism depth displacement
PATH → distance and ocean configuration	OUTCOME → exposure vulnerability and action	PATH → distance and ocean configuration

HAZARD / ACTOR / INSTRUMENT
- SOURCE → mechanism depth displacement
- PATH → distance and ocean configuration

DECISIVE RISK OR MANDATE DISTINCTION
- COAST → bathymetry topography land use
- OUTCOME → exposure vulnerability and action

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- SOURCE → mechanism depth displacement
- PATH → distance and ocean configuration

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COAST → bathymetry topography land use; OUTCOME → exposure vulnerability and action.

03

STAGE 03 DETECTION-CONFIRMATION RAIL

DETECTION-CONFIRMATION RAIL · 1 SEISMIC EVENT ANALYSIS · 2 PRE-RUN MODEL SCENARIO · 3 SEA-LEVEL MONITORING · 4 UPDATED BULLETIN · THREAT STATUS

HAZARD / RISK / RESILIENCE CONCEPT
- 1 SEISMIC EVENT ANALYSIS
- 2 PRE-RUN MODEL SCENARIO

INSTITUTIONAL OR COMMUNITY RESPONSE
- 3 SEA-LEVEL MONITORING
- 4 UPDATED BULLETIN / THREAT STATUS

CAPACITY / STATUS / EVIDENCE LIMIT
- 1 SEISMIC EVENT ANALYSIS
- 2 PRE-RUN MODEL SCENARIO

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 SEA-LEVEL MONITORING; 4 UPDATED BULLETIN / THREAT STATUS.

04

STAGE 04 ITEWC AUTHORITY MAP

ITEWC AUTHORITY MAP · ANALYSIS AND ADVISORY · REGIONAL FRAMEWORK · PROTECTIVE ACTION · LOCAL AUTHORITY · ALL CLEAR

INCOIS / ITEWC → analysis and advisory

IOTWMS / UNESCO-IOC → regional framework → STATE / DISTRICT / LOCAL → protective action → LOCAL AUTHORITY → all clear

HAZARD / RISK / RESILIENCE CONCEPT
- INCOIS / ITEWC → analysis and advisory
- IOTWMS / UNESCO-IOC → regional framework

INSTITUTIONAL OR COMMUNITY RESPONSE
- STATE / DISTRICT / LOCAL → protective action
- LOCAL AUTHORITY → all clear

CAPACITY / STATUS / EVIDENCE LIMIT
- INCOIS / ITEWC → analysis and advisory
- IOTWMS / UNESCO-IOC → regional framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STATE / DISTRICT / LOCAL → protective action.

05

STAGE 05 WARNING-UNCERTAINTY LADDER

WARNING-UNCERTAINTY LADDER · PRELIMINARY INFORMATION · MODEL-BASED THREAT · OBSERVATION-INFORMED UPDATE · THREAT PASSED · LOCAL ALL-CLEAR CHECK

PRELIMINARY INFORMATION

MODEL-BASED THREAT → OBSERVATION-INFORMED UPDATE → FINAL / THREAT PASSED → local all-clear check

RISK CONDITION / INPUT
- PRELIMINARY INFORMATION
- MODEL-BASED THREAT

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- OBSERVATION-INFORMED UPDATE
- FINAL / THREAT PASSED → local all-clear check

LAST-MILE / INCLUSION / OUTCOME LIMIT
- PRELIMINARY INFORMATION
- MODEL-BASED THREAT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FINAL / THREAT PASSED → local all-clear check.

06

STAGE 06 INUNDATION-TO-EVACUATION MAP

INUNDATION-TO-EVACUATION MAP · INUNDATION MAP · ZONE + ROUTE + REFUGE + ASSEMBLY · ACCESSIBLE TRANSPORT + ACCOUNTABILITY · DRILL + FEEDBACK + REVISION

HAZARD / INUNDATION MAP

ZONE + ROUTE + REFUGE + ASSEMBLY → ACCESSIBLE TRANSPORT + ACCOUNTABILITY → DRILL + FEEDBACK + REVISION

RISK CONDITION / INPUT
- HAZARD / INUNDATION MAP
- ZONE + ROUTE + REFUGE + ASSEMBLY

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- ACCESSIBLE TRANSPORT + ACCOUNTABILITY
- DRILL + FEEDBACK + REVISION

LAST-MILE / INCLUSION / OUTCOME LIMIT
- HAZARD / INUNDATION MAP
- ZONE + ROUTE + REFUGE + ASSEMBLY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ACCESSIBLE TRANSPORT + ACCOUNTABILITY; DRILL + FEEDBACK + REVISION.

07

STAGE 07 COASTAL PROTECTION PORTFOLIO

COASTAL PROTECTION PORTFOLIO · SHELTERS WALLS BREAKWATERS WHERE JUSTIFIED · DUNES MANGROVES FORESTS REEFS · CRZ + RISK-SENSITIVE SITING · LAYERED AND LOCATION-SPECIFIC

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
ENGINEERED → shelters walls breakwaters where justified	REGULATORY → CRZ + risk-sensitive siting	ENGINEERED → shelters walls breakwaters where justified
NATURAL → dunes mangroves forests reefs	RULE → layered and location-specific	NATURAL → dunes mangroves forests reefs

HAZARD / ACTOR / INSTRUMENT
- ENGINEERED → shelters walls breakwaters where justified
- NATURAL → dunes mangroves forests reefs

DECISIVE RISK OR MANDATE DISTINCTION
- REGULATORY → CRZ + risk-sensitive siting
- RULE → layered and location-specific

WARNING / IMPLEMENTATION / OUTCOME LIMIT
- ENGINEERED → shelters walls breakwaters where justified
- NATURAL → dunes mangroves forests reefs

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REGULATORY → CRZ + risk-sensitive siting; RULE → layered and location-specific.

08

STAGE 08 COASTAL ECONOMY CONTINUITY

COASTAL ECONOMY CONTINUITY · SHUTDOWN ACCESS REOPENING · BOATS LANDING CENTRES MARKETS · VISITOR WARNING SIGNAGE ROUTE · CONTINUITY AND SAFE RECOVERY

PORT / HARBOUR → shutdown access reopening

FISHERIES → boats landing centres markets → TOURISM → visitor warning signage route → UTILITIES / HOSPITAL → continuity and safe recovery

HAZARD / RISK / RESILIENCE CONCEPT
- PORT / HARBOUR → shutdown access reopening
- FISHERIES → boats landing centres markets

INSTITUTIONAL OR COMMUNITY RESPONSE
- TOURISM → visitor warning signage route
- UTILITIES / HOSPITAL → continuity and safe recovery

CAPACITY / STATUS / EVIDENCE LIMIT
- PORT / HARBOUR → shutdown access reopening
- FISHERIES → boats landing centres markets

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UTILITIES / HOSPITAL → continuity and safe recovery.

09

STAGE 09 TSUNAMI READY CYCLE

TSUNAMI READY CYCLE · MAP RISK · EVACUATION MAP + SIGNAGE · EVALUATE INDICATORS · RENEW READINESS · COMMUNITY RECOGNITION · NOT NATIONAL CERTIFICATION

MAP RISK → EDUCATE → WARN → EVACUATION MAP + SIGNAGE → DRILL → EVALUATE INDICATORS → RENEW READINESS → COMMUNITY RECOGNITION → NOT NATIONAL CERTIFICATION

RISK CONDITION / INPUT
- MAP RISK → EDUCATE → WARN
- EVACUATION MAP + SIGNAGE → DRILL

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM
- EVALUATE INDICATORS → RENEW READINESS
- COMMUNITY RECOGNITION → NOT NATIONAL CERTIFICATION

LAST-MILE / INCLUSION / OUTCOME LIMIT
- MAP RISK → EDUCATE → WARN
- EVACUATION MAP + SIGNAGE → DRILL

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVALUATE INDICATORS → RENEW READINESS; COMMUNITY RECOGNITION → NOT NATIONAL CERTIFICATION.

10

STAGE 10 TRUST UNDER UNCERTAINTY

TRUST UNDER UNCERTAINTY · THREAT POSSIBLE? · ISSUE PRECAUTIONARY MESSAGE · NEW DATA? · EXPLAIN REASON · PRESERVE TRUST · LIMITED IMPACT · NOT AUTOMATIC FALSE-ALARM FAILURE

THREAT POSSIBLE? → ISSUE PRECAUTIONARY MESSAGE

NEW DATA? → UPDATE / UPGRADE / DOWNGRADE → EXPLAIN REASON → PRESERVE TRUST → LIMITED IMPACT → NOT AUTOMATIC FALSE-ALARM FAILURE

HAZARD / RISK / RESILIENCE CONCEPT
- THREAT POSSIBLE? → ISSUE PRECAUTIONARY MESSAGE
- NEW DATA? → UPDATE / UPGRADE / DOWNGRADE

INSTITUTIONAL OR COMMUNITY RESPONSE
- EXPLAIN REASON → PRESERVE TRUST
- LIMITED IMPACT → NOT AUTOMATIC FALSE-ALARM FAILURE

CAPACITY / STATUS / EVIDENCE LIMIT
- THREAT POSSIBLE? → ISSUE PRECAUTIONARY MESSAGE
- NEW DATA? → UPDATE / UPGRADE / DOWNGRADE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NEW DATA? → UPDATE / UPGRADE / DOWNGRADE.

11

STAGE 11 TSUNAMI ANSWER SPINE

TSUNAMI ANSWER SPINE · DEFINE DISPLACEMENT · TRACE SHOALING AND CONSEQUENCE · MAP DETECTION MODEL SEA LEVEL BULLETIN AND AUTHORITY · CONNECT MAP TO EVACUATION BUFFERS PORTS AND LIVELIHOODS · QUALIFY UNCERTAINTY READINESS RECOVERY AND OUTCOME

DEFINE DISPLACEMENT → TRACE SHOALING AND CONSEQUENCE → MAP DETECTION MODEL SEA LEVEL BULLETIN AND AUTHORITY → CONNECT MAP TO EVACUATION BUFFERS PORTS AND LIVELIHOODS → QUALIFY UNCERTAINTY READINESS RECOVERY AND OUTCOME

HAZARD / RISK / RESILIENCE CONCEPT
- DEFINE DISPLACEMENT → TRACE SHOALING AND CONSEQUENCE
- MAP DETECTION MODEL SEA LEVEL BULLETIN AND AUTHORITY

INSTITUTIONAL OR COMMUNITY RESPONSE
- CONNECT MAP TO EVACUATION BUFFERS PORTS AND LIVELIHOODS
- QUALIFY UNCERTAINTY READINESS RECOVERY AND OUTCOME

CAPACITY / STATUS / EVIDENCE LIMIT
- DEFINE DISPLACEMENT → TRACE SHOALING AND CONSEQUENCE
- MAP DETECTION MODEL SEA LEVEL BULLETIN AND AUTHORITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONNECT MAP TO EVACUATION BUFFERS PORTS AND LIVELIHOODS.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · MULTI-AGENCY WARNING ARCHITECTURE · THE INDIAN TSUNAMI EARLY · TECHNO-LEGAL REGIME IS DUAL-PURPOSE · THE CRZ NOTIFICATION · NATURAL-BARRIER EVIDENCE FROM THE 2004 EVENT · VISIONIAS RECORDS · NAVAL HYDROGRAPHIC DEPARTMENT'S DUAL CIVIL-MILITARY ROLE

OPTIONAL RISK / RESILIENCE DEPTH
- Multi-agency warning architecture : the Indian Tsunami Early
- Techno-legal regime is dual-purpose : the CRZ Notification,
- Natural-barrier evidence from the 2004 event : VisionIAS records
- Naval Hydrographic Department's dual civil-military role : seven

MANDATE, WARNING OR OUTCOME LIMIT
- CAUTION: The bathymetry data gap is explicitly named as a limiting
- CAUTION: Physical vulnerability of the sensing network itself :
- CAUTION: VisionIAS names "inadequate community awareness on tsunami risk and
- CAUTION: Structural versus ecosystem-based coastal defence is a genuine.

COMPARATIVE HAZARD / GOVERNANCE NUANCE
- NOT: The "warning system exists, therefore coastal safety is assured"
- CAUTION: Traditional-knowledge documentation gap : VisionIAS names "lack
- CAUTION: VisionIAS does not detail tsunami-specific financing (structural
- Monitoring paleo-tsunami studies and documentation — flagged by

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels